Name: _____

Class: X Section: A

Subject: Biology Date:

Chapter/Topic: Control and Coordination

Delhi Public School
Gurugram | Sector 67A
Nurturing the Whole Child

PART 1: NEURONS, SYNAPSE & REFLEX ARCS

2-Mark Questions

1. Draw the structure of a neuron and label the following parts:
   - (a) Part where information is acquired.
   - (b) Part through which information travels as an electrical impulse
2. What is a synapse? Why is there a brief delay when a signal crosses a synapse?
3. What happens at the synapse between two neurons?
4. Identify the missing components in the following reflex arc pathway.

   Stimulus → Receptor → (A) → Spinal Cord → (B) → Effector → Response
5. Why are signals in a reflex arc processed in the spinal cord rather than the brain?

3-Mark Questions

6. (a) Name the structural and functional unit of the nervous system.
   (b) How does an electrical signal travel within a neuron, and how is it converted into a chemical signal at the axon end?
   (c) What prevents the backward flow of an impulse across a synapse?
7. Differentiate between a Reflex Action and Involuntary Action with one example each.
8. When a person accidentally touches a hot object:
   - (a) Which receptor detects the heat stimulus?
   - (b) Which organ acts as the main integration center for this immediate action?
   - (c) Name the effector tissue that executes the final withdrawal movement.

PART 2: HUMAN BRAIN & PROTECTION OF CNS

2-Mark Questions

9. How is the Central Nervous System (Brain and Spinal Cord) protected against physical shocks and mechanical injuries?
10. A person lost their body balance while walking along a straight line after consuming alcohol. Which specific part of the brain is affected? Mention two other functions of this region.

3-Mark Questions

11. Name the part of the human brain responsible for the following activities:
    - (a) Maintaining posture and equilibrium of the body.
    - (b) Regulating blood pressure, salivation, and vomiting.
    - (c) Sensation of feeling full / hunger.
    - (d) Thinking, intelligence, and voluntary actions.
    - (e) Regulating respiration rate.
    - (f) Controlling reflex movements of the head, neck, and trunk in response to visual/auditory stimuli.

5-Mark Questions

12. (a) Draw a labeled diagram of the Human Brain showing Forebrain, Midbrain, Hindbrain, Cerebrum, Cerebellum, Medulla, and Pituitary gland.
    (b) Explain why an injury to the medulla oblongata can prove immediately fatal to a human being.

PART 3: PLANT HORMONES & TROPIC/NASTIC MOVEMENTS

2-Mark Questions

13. Define Phototropism and Geotropism. Give one diagrammatic example of positive and negative geotropism in a plant.

14. Why do the leaves of the “touch-me-not” plant (*Mimosa pudica*) droop and fold when touched, even though plants have no nervous or muscular tissue?

15. How do tendrils of a pea plant climb around a support or fence? Name the tropic movement involved.

3-Mark Questions

16. Match the plant hormones with their primary physiological functions:
    - (a) Auxin: *Cell elongation and phototropic curvature.*
    - (b) Gibberellins: *Stem elongation and seed germination.*
    - (c) Cytokinins: *Promotes rapid cell division in fruits and seeds.*
    - (d) Abscisic Acid (ABA): *Growth inhibitor, causes leaf wilting and stomatal closure.*

17. Distinguish clearly between Tropic movements and Nastic movements in plants based on:
    - (a) Direction of response relative to stimulus
    - (b) Involvement of growth
    - (c) Speed and reversibility of response

PART 4: HUMAN ENDOCRINE SYSTEM & FEEDBACK MECHANISMS

2-Mark Questions

18. Why are endocrine glands called “ductless glands”? How do their secretions reach target organs?

19. What is emergency hormone? Name the gland that secretes it and list two changes it produces in the human body during a “fight or flight” situation.

3-Mark Questions

20. Name the hormone and corresponding endocrine gland associated with the following conditions:
- (a) Swelling in the neck region (Goitre)
- (b) A boy suddenly growing unusually tall (Gigantism).
- (c) High sugar levels in blood and urine (Diabetes).
- (d) Underdeveloped secondary sexual characteristics in a 16-year-old male.

21. Explain the Feedback Mechanism of hormonal control with the help of a neat schematic diagram showing blood sugar regulation by the pancreas.

5-Mark Questions

22. (a) Compare the modes of action and speed of transmission between the Nervous System and the Endocrine System.

(b) Why is chemical (hormonal) control necessary in multi-cellular organisms alongside electrical impulse control?

[Hint: Electrical impulses do not reach every single body cell, and neurons need time to reset after transmitting an impulse, whereas chemical signals reach all cells via blood and sustain long-term growth/ metabolic changes.]